

From a paper algorithm to Python

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Four packs contain six counters each. What steps find and show the total?

My starting explanation

2. Practise with fading help

Use W2: Predict then observe for From a paper algorithm to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for From a paper algorithm to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace variables and a change

Predict the printed value. Change only packs to 5 and predict again. Explain which statements apply a written rule.

```
packs = 3  
per_pack = 7  
total = packs * per_pack  
print(total)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Connect the paper steps to Python

Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

6. W2 Predict then observe

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual output, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

7. W3 Explain an order error

Move `print(total)` to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

8. W4 Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using `trays`, `per_tray` and `tiles`. Predict its output. Then change only `trays` to 0 and predict again. Explain why adding the two quantities is wrong.

My reasoning and evidence

9. W5 Keep an honest conclusion

Explain the difference between a successful run and a correct calculation. State one limit of this program and what you would keep so another person can check your work.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled from a paper algorithm to python evidence sheet to Python readiness portfolio. Show how you can translate a short paper calculation into a python script. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: From a paper algorithm to Python

Convert a seed-count algorithm. The paper plan counts seeds in identical trays, then adds a labelled spare tray.

Python 3: `trays=4; per_tray=6; spare=2; total=trays*per_tray+spare; print(total)`. All quantities count seeds.

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From a paper algorithm to Python

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the inputs

Four trays hold six seeds each; two extra seeds are separate.



2. Translate the plan

Multiply tray count by seeds per tray, then add the spare count.



3. Trace the assignment

$4 \times 6 + 2$ is 26; total stores 26 and print displays it.

CHANGE ONE THING

Change spare to 5 while keeping both tray inputs fixed.

Explain the changed result using the matching lesson.

Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.